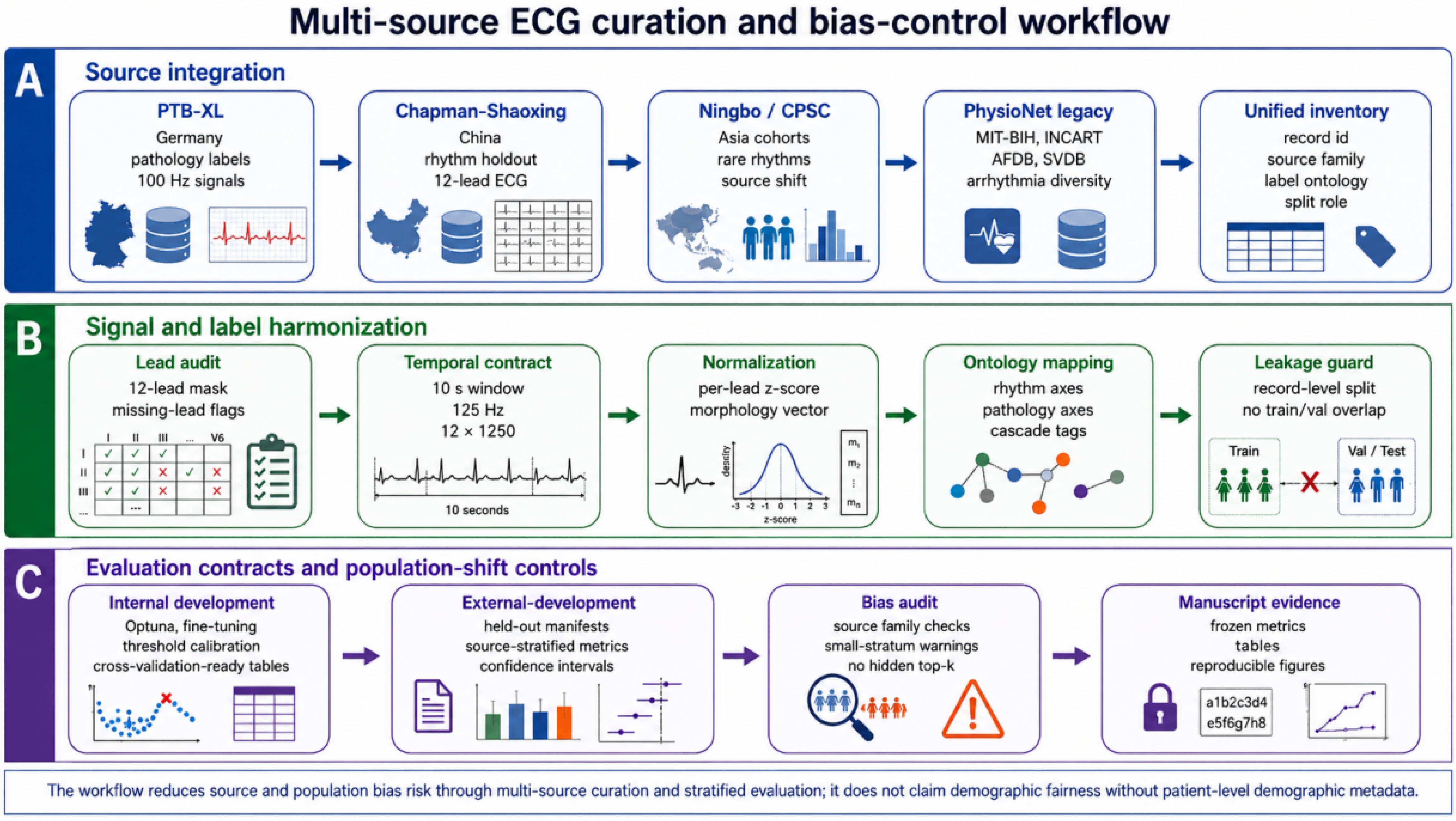
Multi-source ECG curation and bias-control workflow
A
Source integration
PTB-XL
Germany
pathology labels
100 Hz signals
Chapman-Shaoxing
China
rhythm holdout
12-lead ECG
Ningbo / CPSC
Asia cohorts
rare rhythms
source shift
PhysioNet legacy
MIT-BIH, INCART
AFDB, SVDB
arrhythmia diversity
Unified inventory
record id
source family
label ontology
split role
B
Signal and label harmonization
Lead audit
12-lead mask
missing-lead flags
I
II
III
...
V6
Temporal contract
10 s window
125 Hz
12 × 1250
10 seconds
Normalization
per-lead z-score
morphology vector
density
z-score
Ontology mapping
rhythm axes
pathology axes
cascade tags
Leakage guard
record-level split
no train/val overlap
Train
Val / Test
C
Evaluation contracts and population-shift controls
Internal development
Optuna, fine-tuning
threshold calibration
cross-validation-ready tables
External-development
held-out manifests
source-stratified metrics
confidence intervals
Bias audit
source family checks
small-stratum warnings
no hidden top-k
Manuscript evidence
frozen metrics
tables
reproducible figures
a1b2c3d4
e5f6g7h8
The workflow reduces source and population bias risk through multi-source curation and stratified evaluation; it does not claim demographic fairness without patient-level demographic metadata.